

Sentiment and Opinion

- ❑ **Sentiment:** a view of attitude/emotion toward a situation or event.
- ❑ **Opinion:** a view or judgement formed about something.
- ❑ Sentiment and opinion are heavily correlated.
 - ❑ Note: many opinions may express sentiment, but some do not. (e.g., “I think she will move to Canada after graduation.”)
- ❑ Subtle difference between sentiment and opinion:
 - ❑ Sentiment sentence example: “I find data mining highly interesting.”
 - ❑ Opinion sentence example: “I believe data mining is promising and useful.”
 - ❑ The subtle difference is that one may share the same sentiment expressed in the example of sentiment sentence and agree or disagree with the opinion in the second sentence.

Sentiment Analysis and Opinion Mining Techniques

- ❑ Goal: understand sentiment expressed in text.
- ❑ Techniques being used:
 - ❑ Natural language processing methods.
 - ❑ Machine learning and data mining methods.
 - ❑ Psychology and social sciences.
- ❑ Technical directions being explored:
 - ❑ Analysis at different levels: Document-level, sentence-level, aspect-level.
 - ❑ Sentiment classification based on lexicons.
 - ❑ Two types of text content: *stand-alone* and *online conversations*.
 - ❑ Mining intent: intent may imply sentiments and express opinions.
 - ❑ Opinions spam detection: acquire data for social goodness.

Sentiment Analysis and Opinion Mining Applications

- ❑ Understand advantages/disadvantages of targets:
 - ❑ Sentiment analysis/opinion mining over product reviews.
 - ❑ Help customer make decisions; Help platform optimize supply chain; Help producers and manufacturers improve products, etc.
- ❑ Outcome can be consumed by other AI agents:
 - ❑ Example: sentiment from news ➡ trading signals for stock markets.
 - ❑ Challenges: highly efficient/accurate/robust system.
- ❑ Monitoring and government administration:
 - ❑ Fight against cyber-violence, terrorism and racism.
 - ❑ Challenges: fairness and privacy issues.